

Dispute Triage: First Response Within SLA

A Rule-Based and Data-Driven Prioritisation Framework for UPI & Card Dispute Management in Indian Banking

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Introduction: The Challenge of Dispute Management

Indian banks today face an unprecedented surge in UPI and card disputes, driven by the exponential growth of digital transactions. The volume and complexity of these disputes have created significant operational pressures, affecting both customer satisfaction and regulatory compliance.

The consequences of inefficient dispute handling extend far beyond operational metrics. When banks fail to respond promptly and effectively, they face multiple critical challenges that compound over time.



First Response Delays

Critical disputes languish without initial acknowledgement, creating customer anxiety and regulatory exposure



Misprioritised Cases

High-value and fraudulent transactions receive the same attention as routine queries, leading to significant financial risks



Processing Bottlenecks

Cases become trapped in intermediate stages, with no mechanism to identify or escalate aged disputes



Information Gaps

Customers receive minimal updates during processing, creating frustration and repeat complaints



SLA Violations

Regulatory deadlines are breached systematically, exposing banks to penalties and reputational damage



This project proposes a comprehensive triage and prioritisation system that fundamentally transforms dispute management. By implementing intelligent routing, automated scoring, and proactive communication, the framework ensures that every dispute receives appropriate attention within regulatory timelines whilst maintaining full compliance with RBI dispute-resolution standards.

Business Context: Understanding the Dispute Ecosystem

The current dispute resolution landscape in Indian banking is characterised by significant complexity, involving multiple stakeholders and intricate inter-bank coordination mechanisms. When customers experience UPI or card transaction failures, their complaints traverse a sophisticated sequence of stages, each requiring specific actions, documentation, and inter-institutional communication.

The Complaint Journey

Complaints move through a carefully orchestrated sequence of stages, from initial logging through review, system processing, and ultimately to reversal or resolution. Analysis of historical data reveals that delays are concentrated primarily in mid-to-late lifecycle stages rather than during initial registration, indicating that the bottleneck lies not in complaint intake but in inter-bank coordination and settlement processes.

Each stage requires coordination amongst multiple entities, creating natural points of delay where manual intervention, documentation verification, or external system responses are required. Understanding these delay patterns is critical to designing an effective triage mechanism.

Multiple Stakeholders

A single complaint can involve coordination amongst several banking entities, each with distinct responsibilities and processing timelines:

- **Issuing Bank:** The customer's bank that initiated or authorised the original transaction
- **Acquiring Bank:** The merchant's bank that processes payment acceptance
- **NPCI:** National Payments Corporation of India, managing the UPI infrastructure
- **Merchant Bank:** The entity maintaining merchant accounts and settlement
- **Core Banking Systems:** Internal platforms managing account updates and reconciliation

❏ **Critical Insight:** The involvement of multiple entities creates natural friction points in the dispute lifecycle. A structured triage system is essential not just for efficiency, but to avoid cascading backlogs during peak transaction periods such as festival seasons, when dispute volumes can surge by 200-300%.

Types of Disputes: Categorising Customer Complaints

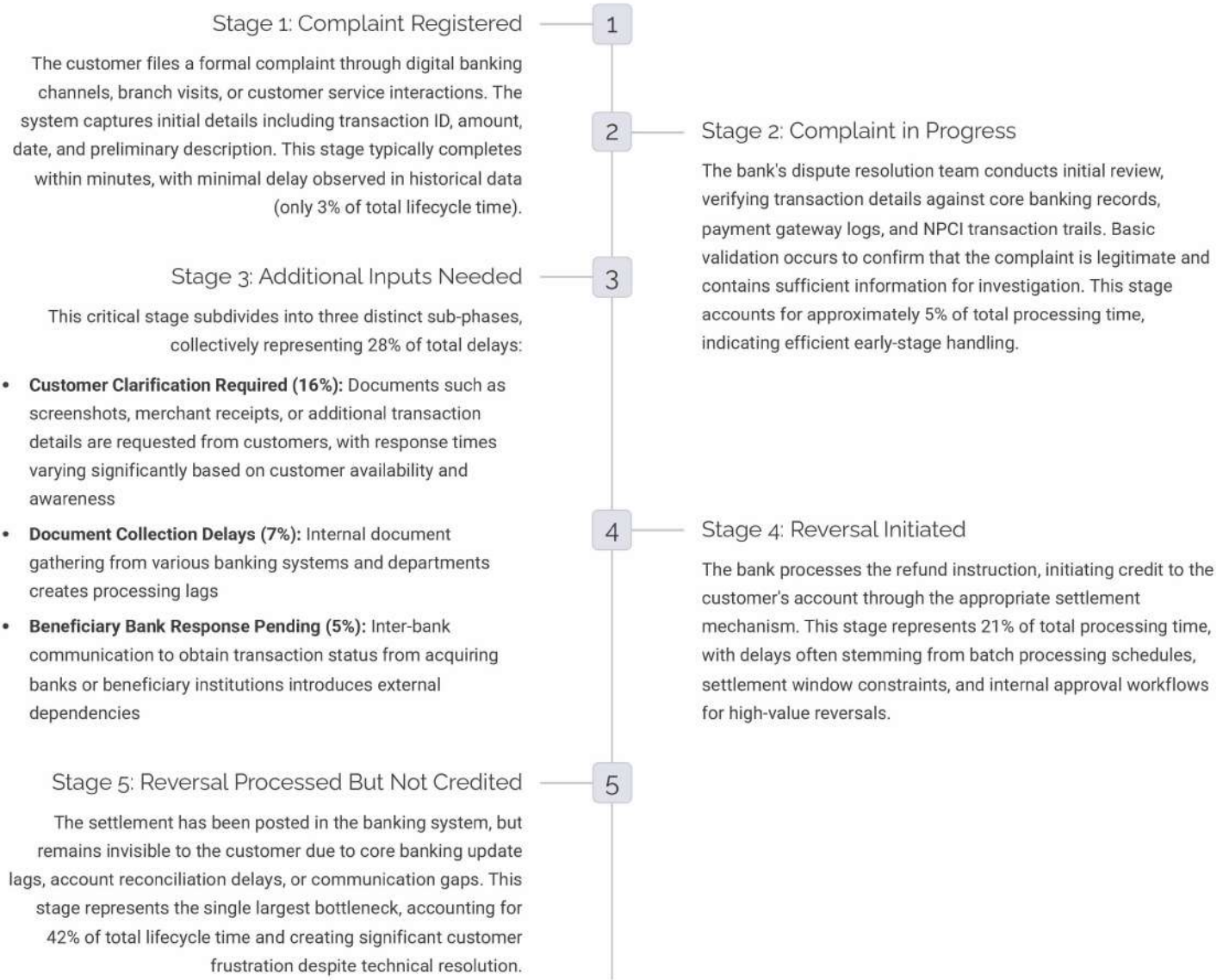
Analysis of bank data reveals seven distinct dispute categories, each characterised by unique risk profiles, urgency levels, and customer impact. Understanding these categories is fundamental to designing an effective triage framework, as different dispute types require varying levels of investigation, documentation, and inter-bank coordination.

Amount Debited but Not Credited Risk Level: High The most common dispute type where funds are deducted from customer accounts but fail to reach the intended recipient. Requires urgent investigation to determine whether the failure occurred at the issuing bank, acquiring bank, or within the NPCI network. Customer impact is immediate and significant, particularly for high-value transactions.	Amount Debited – Transaction Timeout Risk Level: Medium Occurs when network latency or system unavailability causes transaction failures after debit initiation. These cases often self-resolve through automated reconciliation within 24-48 hours, but require monitoring to ensure proper reversal. Customer anxiety is high despite the technical nature of the issue.	Reversal Initiated but Not Reflected Risk Level: Medium-High The bank has processed the refund, but it remains invisible in the customer's account due to settlement delays or core banking system update lags. These cases create significant customer frustration as the resolution is technically complete but not visible to the end user.
Wrong Beneficiary Transfer Risk Level: Critical Customer has inadvertently transferred funds to an incorrect recipient, often due to mistyped UPI IDs or account numbers. Resolution requires beneficiary cooperation and can involve legal considerations if the recipient refuses to return funds. Immediate escalation is essential.	Transaction Not Initiated by Customer (Fraud) Risk Level: Critical Unauthorised transactions indicating potential account compromise, credential theft, or social engineering attacks. These cases require immediate investigation, account protection measures, and coordination with fraud prevention teams. Regulatory reporting may be mandatory.	Duplicate Transactions Risk Level: Medium Multiple debits for a single intended transaction, typically caused by customer interface errors or system glitches during retry attempts. Generally straightforward to resolve through transaction log analysis, but requires prompt action to maintain customer trust.
Merchant Not Received Payment Risk Level: Medium Merchant claims non-receipt of payment despite customer debit. Requires coordination with acquiring banks and merchant banking partners. Resolution complexity varies based on whether the issue lies in settlement timing or genuine payment routing failures.		

Each dispute type varies significantly in terms of risk exposure, urgency requirements, and customer impact. A sophisticated triage system must automatically recognise these categories and assign appropriate priority levels, ensuring that critical cases such as fraud and wrong beneficiary transfers receive immediate attention whilst routine timeout issues follow standard processing protocols.

Existing Dispute Lifecycle: Five Critical Stages

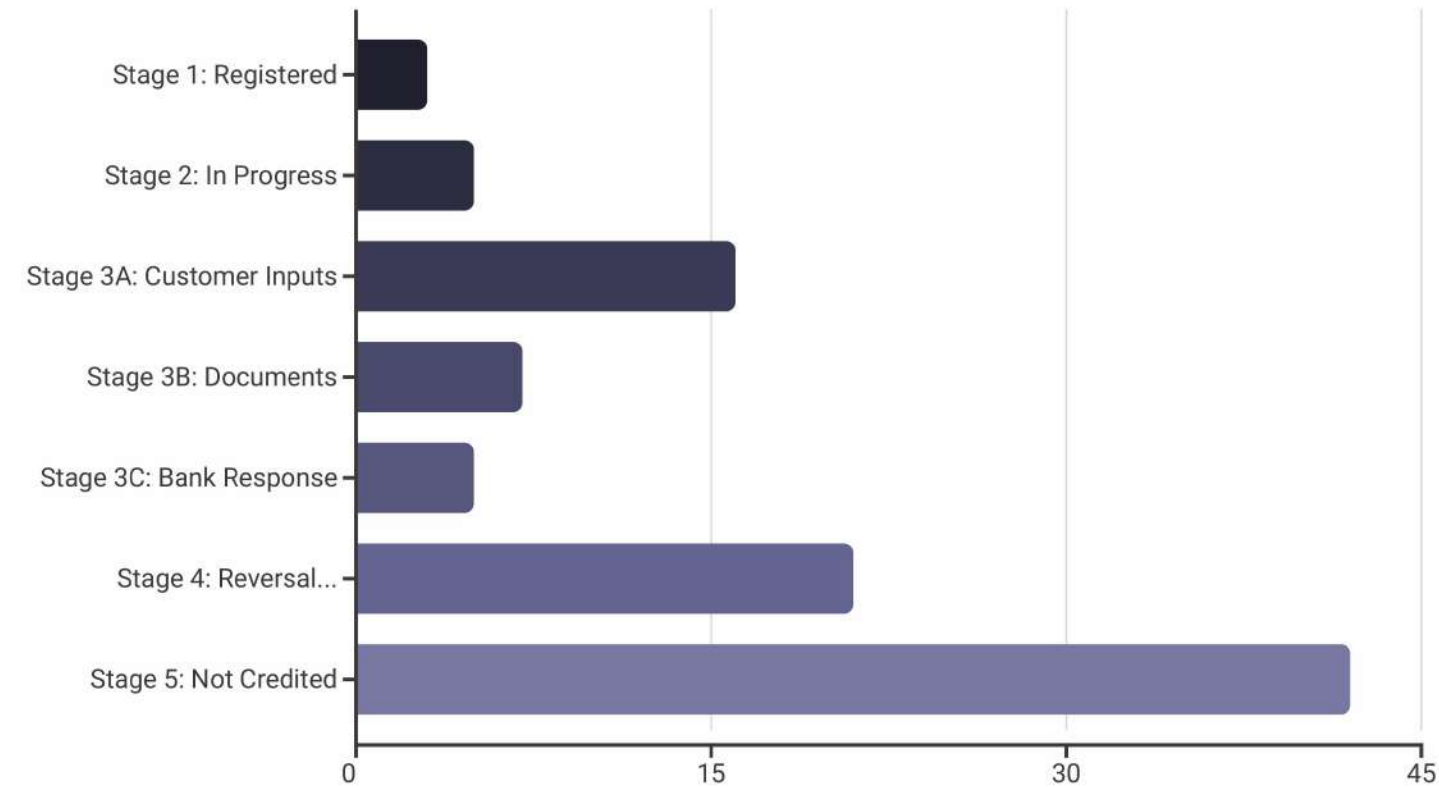
The current dispute resolution process follows a structured five-stage lifecycle, each representing a distinct phase in the journey from complaint registration to final resolution. Understanding the characteristics, requirements, and typical delays at each stage is essential for identifying intervention points where triage logic can optimise processing efficiency.



❑ **Key Observation:** Delays are heavily concentrated in the latter stages of the lifecycle, with Stages 4 and 5 alone accounting for 63% of total processing time. This indicates that the primary challenge is not complaint intake or initial review, but rather the settlement and communication mechanisms that finalise dispute resolution.

Stage-Wise Delay Analysis: Quantifying the Bottleneck

Detailed analysis of historical dispute data reveals striking patterns in delay distribution across the five lifecycle stages. Understanding these patterns is crucial for designing targeted interventions and prioritisation rules that address the most significant bottlenecks.



Critical Insights from Delay Data

- 1 Post-Reversal Bottleneck Dominates**

58% of all disputes experience delays after reversal initiation (Stages 4 and 5 combined), indicating that the primary challenge lies not in investigation or decision-making, but in settlement execution and visibility.
- 2 Documentation Gaps Are Significant**

Stage 3 contributes 28% of total delays across its three sub-phases, with customer clarification requests (16%) being the largest single component. This suggests opportunities for automation in document collection and customer communication.
- 3 Early Stages Function Efficiently**

Registration and initial review (Stages 1 and 2) account for only 8% of delays combined, demonstrating that the complaint intake pipeline is functioning effectively and does not require triage intervention.

Implications for Triage Design

- The data reveals that conventional approaches focusing solely on complaint registration and initial routing are insufficient. An effective triage system must:
- Prioritise Stage 5 Cases:** Disputes stuck after reversal processing require bulk prioritisation and proactive communication to customers about settlement timelines
 - Automate Stage 3 Document Collection:** Implementing pre-defined document checklists and automated customer communication can reduce the 28% delay contribution from this stage
 - Distinguish Critical from Routine Cases:** Currently, no mechanism exists to differentiate high-risk, high-value disputes from routine timeout issues, leading to uniform processing that under-serves critical cases
 - Address Inter-Bank Coordination:** The 5% delay from beneficiary bank responses requires improved tracking and escalation mechanisms for external dependencies
- The triage system must therefore address mid-stage operational delays and post-reversal settlement bottlenecks, not merely optimise initial complaint entry and routing.

Root Causes: Understanding System Failures

Comprehensive analysis of the current dispute management process reveals three interconnected categories of failures: operational challenges, customer experience deficiencies, and systemic technology limitations. Addressing these root causes requires a holistic redesign of prioritisation logic and workflow automation.

Operational Challenges

- **Uniform Priority Assignment:** All complaints receive identical initial priority regardless of amount, fraud risk, or customer segment, creating a first-in-first-out processing model that systematically under-serves critical cases
- **Manual Case Sorting:** Operations teams spend significant time manually reviewing and re-prioritising cases that should have been automatically flagged during intake, diverting resources from actual resolution work
- **Insufficient Tracking of Aged Disputes:** No automated mechanism identifies cases approaching SLA deadlines, leading to reactive fire-fighting rather than proactive management
- **High Dependency on Inter-Bank Responses:** External dependencies on acquiring banks and NPCI create delays that are not tracked or escalated systematically
- **Resource Allocation Inefficiencies:** Senior specialists handle routine timeout cases whilst junior staff may inadvertently process fraud complaints, inverting the optimal skill-to-case matching

Customer Experience Issues

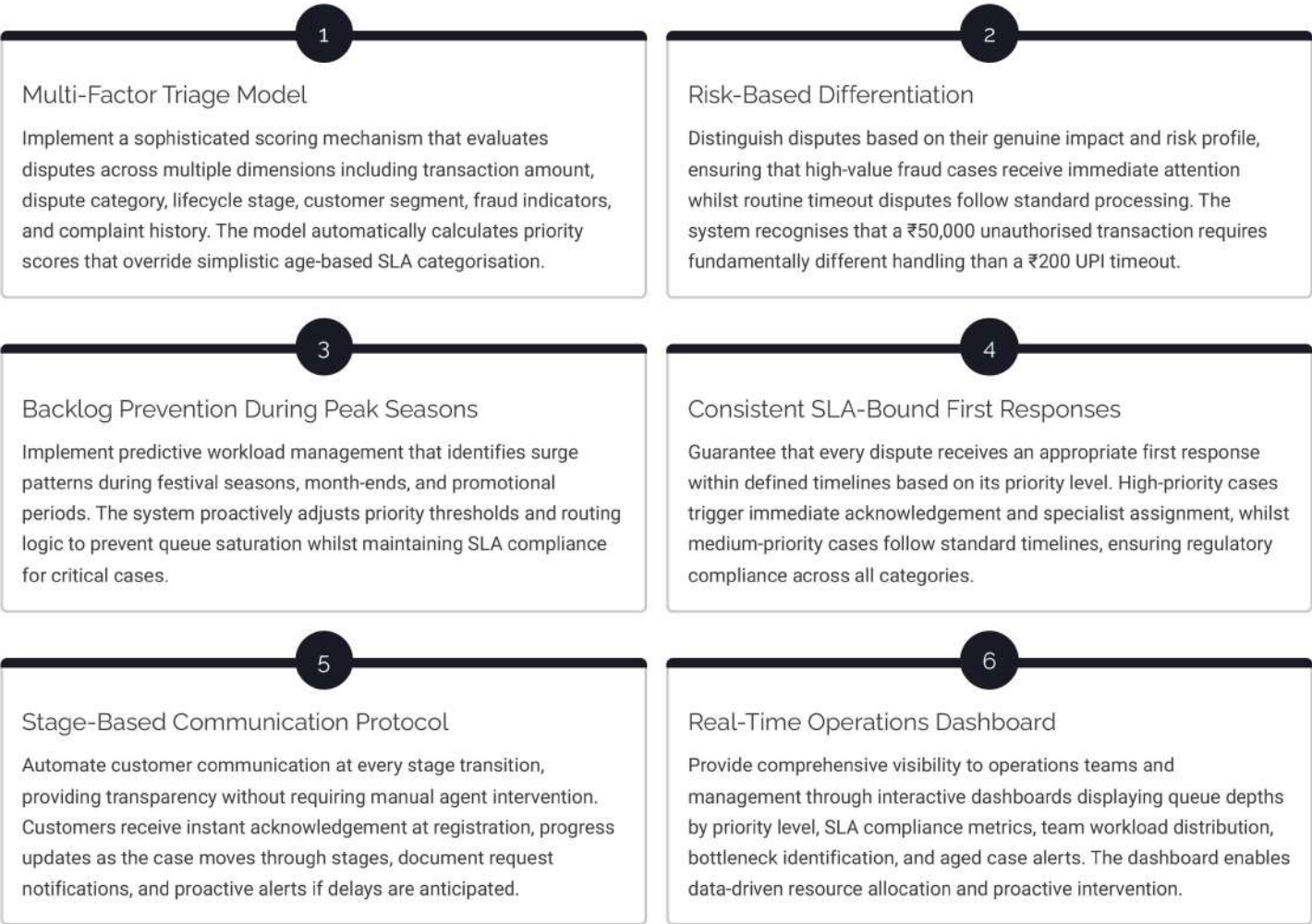
- **Lack of Transparency:** Customers receive minimal visibility into complaint status, leading to anxiety and repeat follow-up calls that further burden operations teams
- **No Proactive Updates:** Communication occurs only when customers initiate contact, rather than at natural stage transitions or when delays are anticipated
- **Undifferentiated Treatment:** High-value customers experiencing fraud receive the same communication and attention as routine low-value timeout cases, damaging premium customer relationships
- **Unclear Timelines:** Customers are provided generic SLA commitments rather than case-specific estimates based on dispute type and complexity
- **Resolution Ambiguity:** Even after reversal processing, customers remain uncertain about when funds will be visible in their accounts, generating unnecessary complaints

System and Technology Issues

- **No Automated Triage Engine:** Absence of rule-based or AI-driven prioritisation means every case requires manual review before appropriate routing
- **SLA Based Only on Age:** Current SLA categorisation considers only days elapsed, ignoring case severity, amount, fraud indicators, and customer segment importance
- **No Alerting System for SLA Breaches:** Teams discover deadline violations after the fact rather than receiving advance warnings that enable preventive action
- **Fragmented Data Sources:** Complaint details, transaction logs, and settlement status reside in separate systems, requiring manual correlation
- **Limited Dashboard Capabilities:** Management lacks real-time visibility into queue depths, team workload distribution, and bottleneck identification
- **Manual Communication Triggers:** Customer updates require individual agent action rather than automated stage-based notifications

Project Objectives: A Comprehensive Triage Framework

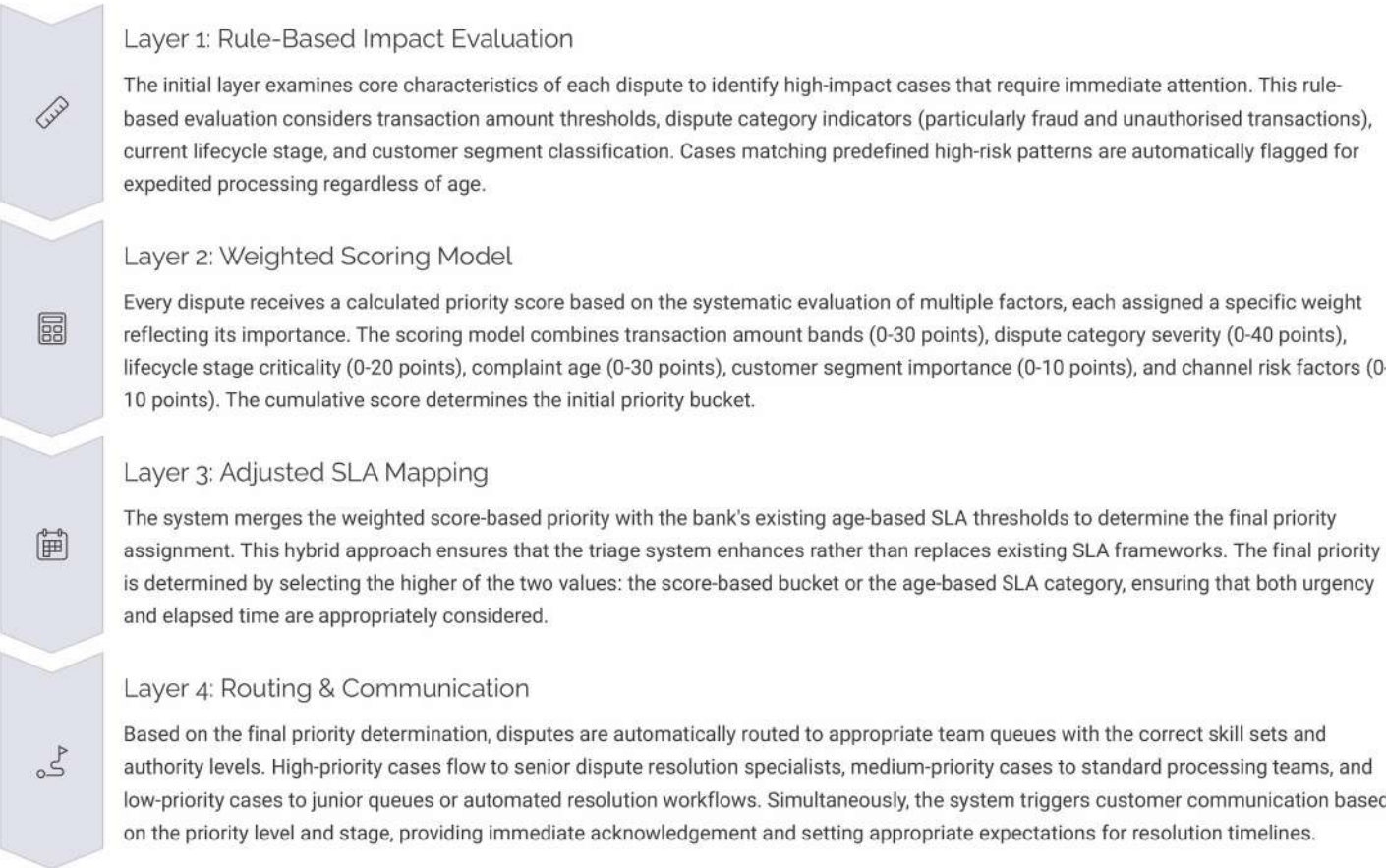
The proposed solution establishes a complete transformation of dispute management through intelligent automation, multi-factor prioritisation, and proactive customer communication. The framework is designed to address the identified root causes whilst maintaining full compliance with RBI regulations and existing bank operational structures.



❑ **Strategic Alignment:** These objectives directly address the root causes identified in the previous analysis whilst maintaining compatibility with existing bank systems and regulatory requirements. The framework is designed for phased implementation, allowing operational teams to adapt gradually whilst delivering immediate value from early-stage automation.

Proposed Triage Framework: Multi-Layer Intelligence

The triage framework employs a sophisticated four-layer architecture that combines rule-based evaluation, weighted scoring, adjusted SLA mapping, and automated routing. This multi-layer approach ensures that prioritisation decisions reflect genuine business impact whilst maintaining operational efficiency and regulatory compliance.



Key Advantages

- Automated prioritisation reduces manual review time by 70-80%
- Multi-factor scoring captures nuanced risk profiles that age-based SLAs miss
- Hybrid approach ensures continuity with existing SLA frameworks
- Transparent scoring enables audit trails for regulatory compliance
- Flexible weights allow calibration based on bank-specific priorities

Implementation Considerations

- Weight values require initial calibration using historical dispute data
- Regular review cycles ensure scoring remains aligned with evolving risk patterns
- Exception handling protocols address edge cases not captured by standard rules
- Integration with existing CRM and ticketing systems is essential
- Change management for operations teams adapting to automated routing

Scoring Parameters and Case Examples

Comprehensive Weight Assignment

Parameter	Condition	Weight
Transaction Amount	>₹10,000	30
Transaction Amount	₹2,000-10,000	15
Transaction Amount	<₹2,000	0
Dispute Category	Fraud/Unauthorised	40
Dispute Category	Wrong Beneficiary	25
Dispute Category	Not Credited	20
Dispute Category	Timeout/Technical	10
Lifecycle Stage	Stage 5 (Not Credited)	20
Lifecycle Stage	Stage 4 (Reversal Initiated)	10
Lifecycle Stage	Stage 3 (Documentation)	5
Days Open	>25 days	30
Days Open	12-25 days	20
Days Open	4-11 days	10
Customer Segment	Premium/Corporate	10
Channel Risk	Card Transaction	10
Complaint History	High Frequency	5

Priority Mapping: Score >70 = HIGH | Score 41-70 = MEDIUM | Score ≤40 = LOW

The scoring system demonstrates how identical complaint ages receive dramatically different priority treatments based on risk factors. A one-day-old fraud complaint correctly receives immediate attention, whilst a one-day-old low-value timeout follows standard processing, optimising resource allocation without compromising critical cases.

Real-World Case Examples

Case #1: Critical Fraud Alert

Details: ₹37,500 | Unauthorised Transaction | Day 1 | Stage 1

Scoring:

- Amount >₹10k: +30
- Unauthorised: +40
- Days: 0
- Stage: 0

Total: 70 → HIGH Priority

Routing: Senior Dispute Resolution Team

SLA: First response within 2 hours, fraud investigation initiated immediately

Case #2: Medium Priority Reversal

Details: ₹4,200 | Reversal Initiated | Day 8 | Stage 4

Scoring:

- Amount ₹2-10k: +15
- Stage 4: +10
- Days 4-11: +10
- Category: +10

Total: 45 → MEDIUM Priority

Routing: Standard Processing Team

SLA: Settlement tracking within 24 hours

Case #3: Low Priority Timeout

Details: ₹150 | UPI Timeout | Day 0 | Stage 1

Scoring:

- Amount <₹2k: 0
- Timeout: +10
- Days: 0
- Stage: 0

Total: 10 → LOW Priority

Routing: Automated resolution queue with 24-hour reconciliation check

SLA: Auto-acknowledgement, 3-day resolution window